MA²RAG Multi-Agent Experiments

Detailed Architectures and Final Results

Final systems only: **E4**, **M1v8**, **M2 DRHR**, **M3 CEP**, **M4v3 OSPREY**

Pradyut Nair · MSc Thesis · UvA MultIX · Feb 2026

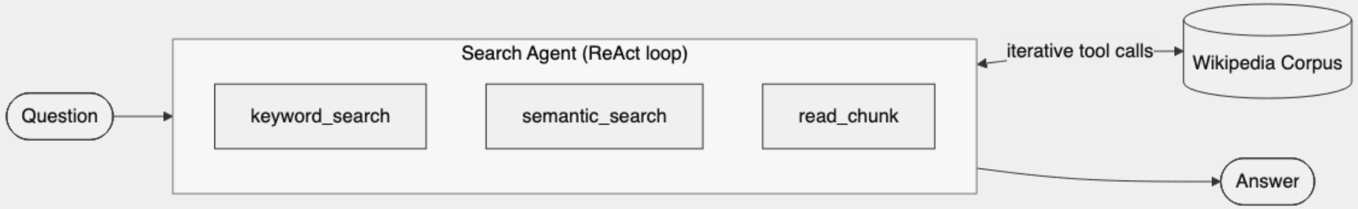
Shared Setup and Final Scoreboard

- Model stack: Qwen3-30B-A3B + E5-base-v2
- Judge: DeepSeek-R1-Distill-Qwen-32B
- Data: 1000 questions each on HotpotQA, 2WikiMultihop, MuSiQue
- Shared tools: `keyword_search`, `semantic_search`, `read_chunk`, `finish`

System	HotpotQA	2WikiMH	MuSiQue	Mean
E4 (single-agent baseline)	66.5	56.9	37.6	**53.7**
M1v8	54.9	32.3	30.3	39.2
M2 DRHR	63.4	34.2	27.2	41.6
M3 CEP	63.4	35.4	28.7	42.5
M4v3 OSPREY	**63.8**	**44.0**	**32.2**	**46.7**

Trend: M1 -> M2 -> M3 -> M4 improves mean from 39.2 to 46.7, but E4 remains highest at 53.7.

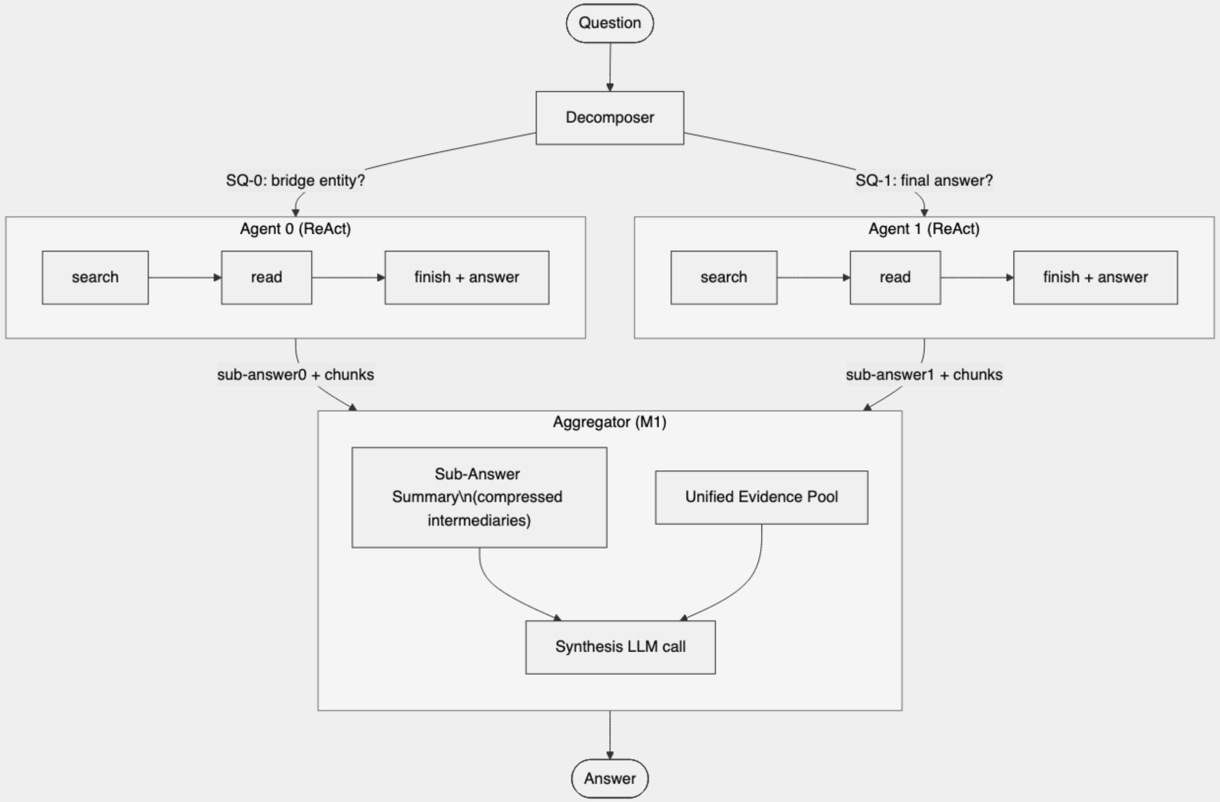
E4 Baseline: Single-Agent A-RAG



- One ReAct agent handles full question end-to-end.
- Iterative loop: search -> read -> decide -> `finish(answer=...)`.
- No decomposition, no wave scheduling, no cross-agent coordination.
- Typical usage: ~2.66-3.05 loops, ~800-873 retrieved tokens.

Dataset	LLM-Acc
HotpotQA	66.5
2WikiMH	56.9
MuSiQue	37.6
Mean	**53.7**

M1v8: Sub-Answer Aggregation

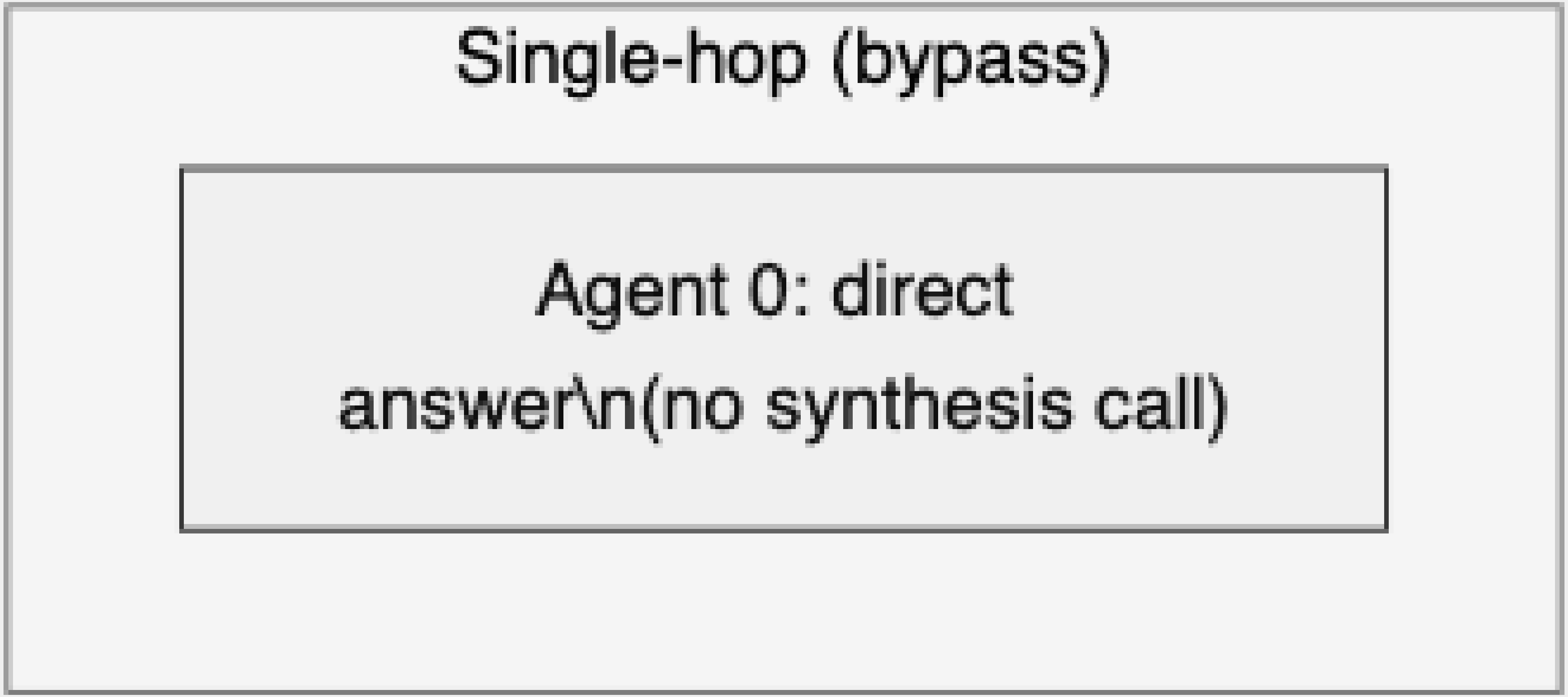


- Decomposer classifies `comparison` / `bridge` / `single_hop`.
- Sub-questions are solved by ReAct agents.
- Aggregator uses both sub-answers and retrieved chunks.
- In bridge chains, wrong SQ-0 can corrupt SQ-1.
- v8 specifics: comparison instruction fixed, self-verify disabled.

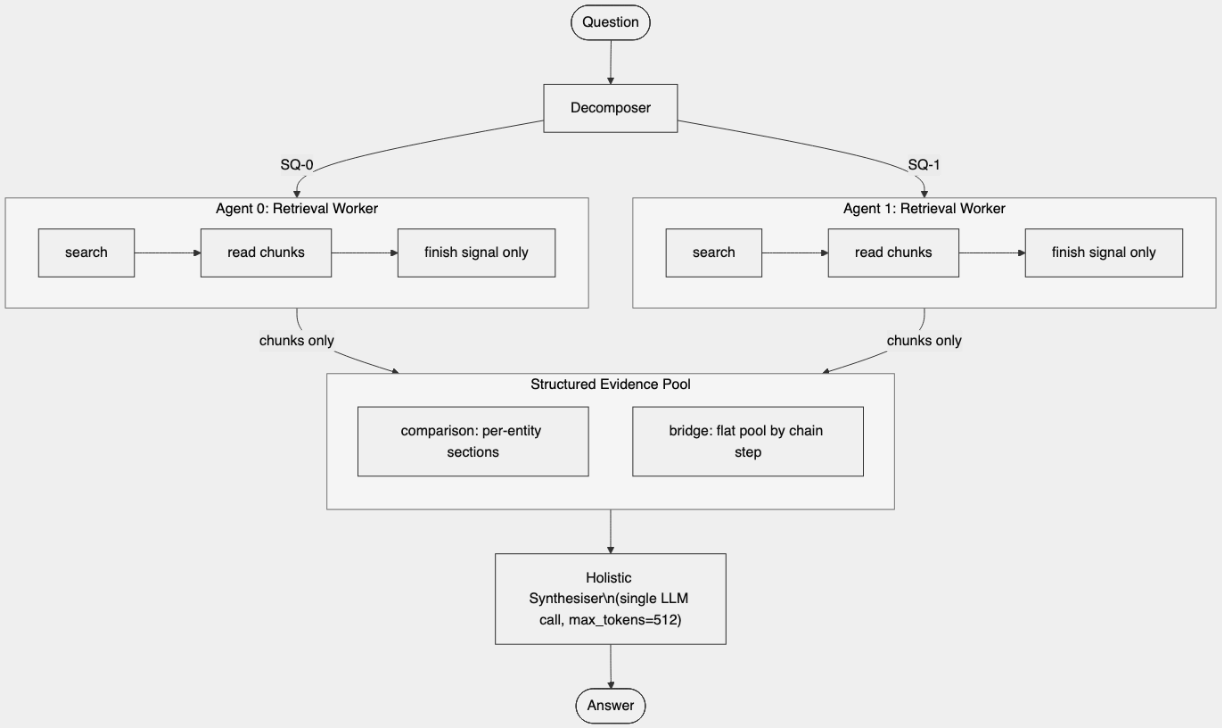
Dataset	LLM-Acc
HotpotQA	54.9
2WikiMH	32.3
MuSiQue	30.3
Mean	**39.2**

Limitation: sub-answer compression loses evidence and amplifies early-hop errors.

Dispatch Strategy (M1/M2/M3)



M2 DRHR: Decomposed Retrieval, Holistic Reasoning

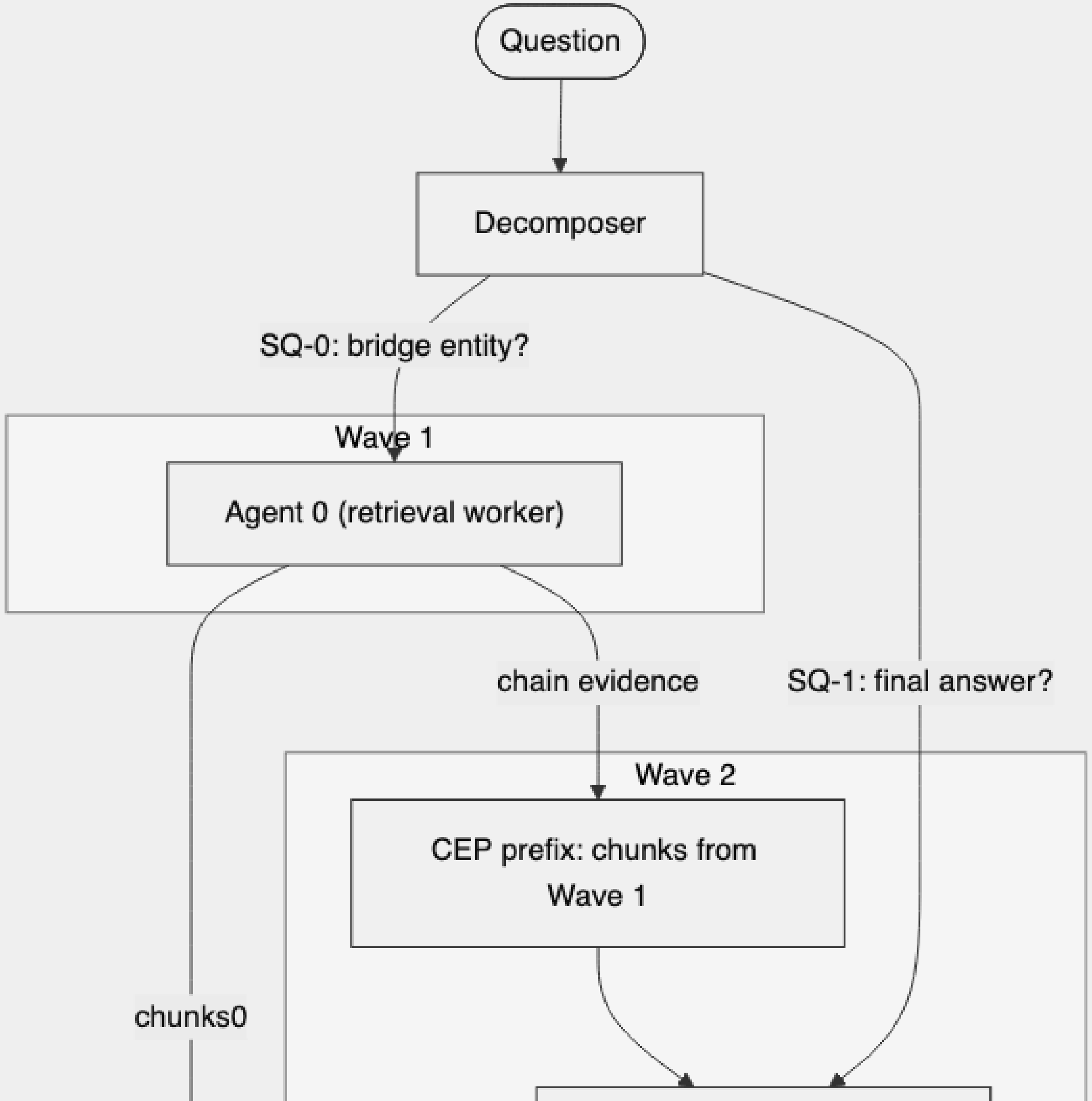


- Same decomposition + dispatch as M1.
- Agents become retrieval workers; sub-answers are ignored.
- Aggregator performs one holistic synthesis on raw chunks only.
- Pooling policy: comparison per-entity sections, bridge flat chain-ordered pool.
- Synthesis cap set to `max_tokens=512`.

Dataset	LLM-Acc
HotpotQA	63.4
2WikiMH	34.2
MuSiQue	27.2
Mean	**41.6**

Effect: big recovery on HotpotQA vs M1 (+8.5), but bridge bottleneck remains on 2WikiMH.

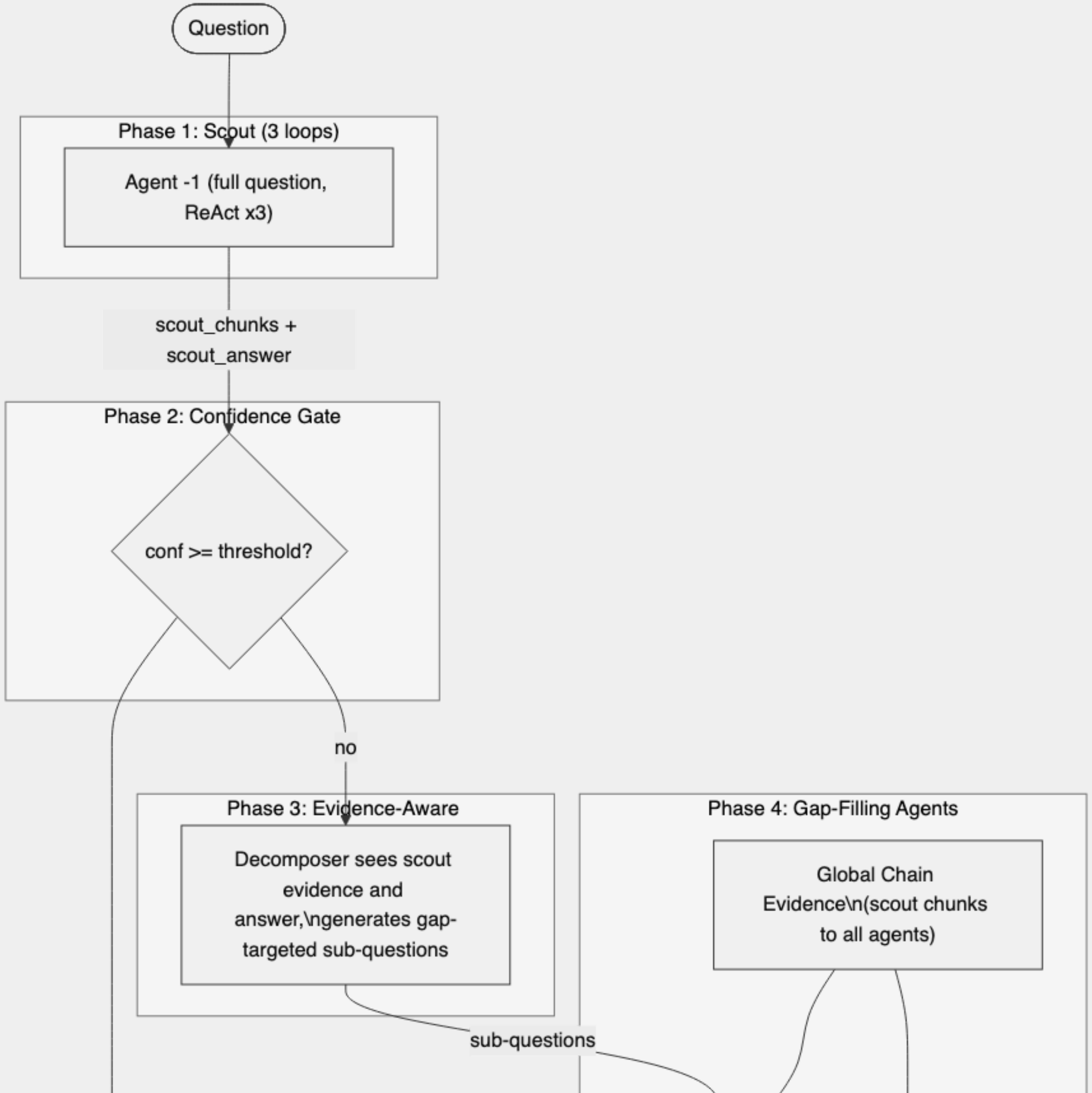
M3 CEP: Chain Evidence Propagation



- M2 + CEP for bridge questions.
- Wave `k` agents receive read-only chunks from waves `0..k−1`.
- CEP prompt explicitly asks agents to use prior evidence to find missing intermediate entity.
- Comparison and single-hop behavior unchanged from M2.

Dataset	LLM-Acc
HotpotQA	63.4
2WikiMH	35.4
MuSiQue	28.7
Mean	**42.5**

M4v3 OSPREY: Scout + Evidence-Aware Pipeline



- 5 phases: Scout -> Gate -> Evidence-aware decompose -> Gap-fill agents -> Anchored synthesis.
- v3 disables fast exit (`threshold=1.1`), so all samples run full evidence path.
- Scout uses agent index `-1` (sentinel), not treated as sub-question agent.
- Scout chunks are injected globally; bridge also gets CEP wave evidence.
- Synthesis prepends `[Doc | Scout]` before sub-question chunks.

Dataset	LLM-Acc
HotpotQA	63.8
2WikiMH	44.0
MuSiQue	32.2
Mean	**46.7**

Comparative Results: Gap to E4 and OSPREY Breakdown

System	HotpotQA gap	2WikiMH gap	MuSiQue gap	Mean gap
M2 DRHR	-3.1	-22.7	-10.4	-12.1
M3 CEP	-3.1	-21.5	-8.9	-11.2
M4v3 OSPREY	**-2.7**	**-12.9**	**-5.4**	**-7.0**

M4v3 by question type	bridge	comparison	single_hop	Overall
HotpotQA	64.2	62.7	63.5	63.8
2WikiMH	30.7	**63.0**	18.8	44.0
MuSiQue	32.3	12.0	35.2	32.2

Interpretation: OSPREY is strong on 2Wiki comparison, but bridge and implicit single-hop inference remain core deficits.

Token Cost and Failure Taxonomy

M4v3 token overhead	Scout avg	Phase-2 agents	Aggregator	Total avg
HotpotQA	3,293	9,306	4,171	12,599
2WikiMH	4,116	12,223	5,521	16,339
MuSiQue	4,549	16,092	6,224	20,641

- Scout overhead is ~25-30% of total M4v3 tokens.

MuSiQue E4 failure taxonomy	% failures
Retrieved but couldn't synthesize	58.8
Searched but missed	17.8
Never searched hop-2	12.8
Decomposition failure	10.1
Corpus gap	0.5

Next target: bridge-aware decomposition constraints + verifier-guided re-query to reduce synthesis and missed-hop failures.

Sources: `02-arag-multi-agent/README.md`, `02-arag-multi-agent/results/*/predictions_eval_summary.json`, `01-arag-reproduction/results/README.md`.